UESTCHN3007 Dynamics & Control
Lecture 1

Overview of Dynamic Systems

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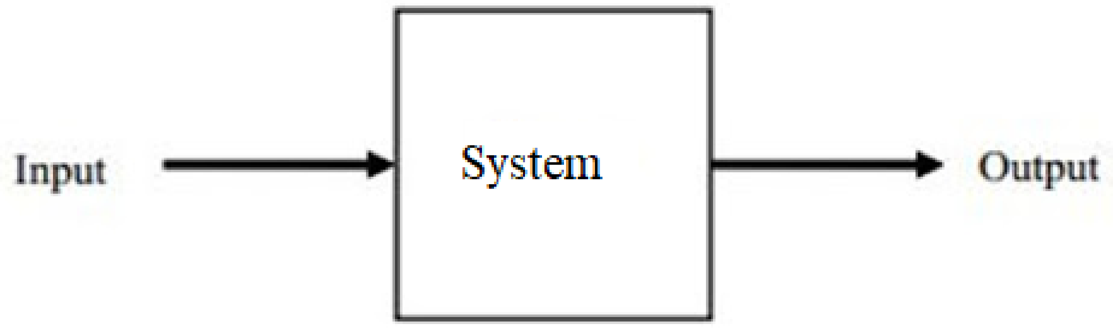
Systems

- Sets of connected objects or things
- They can be living things
- They can be mechanical things
- Behaviors of these systems are shaped by many factors
- All these systems exhibit some common behavior patterns

What is a Dynamic System?

- A dynamic system can be set in motion by applying an input (or excitation) or by disturbing it from its equilibrium.
- The motion of the system is described by its output (or response).
- The output of the system is dependent on the input and the behaviour of the system itself.
- If we know what the behaviour of the system is, we can specify an input so that we get a desired output.
- But how do we describe the behaviour of the system?

The output of the system is dependent on the input and the behaviour of the system itself.

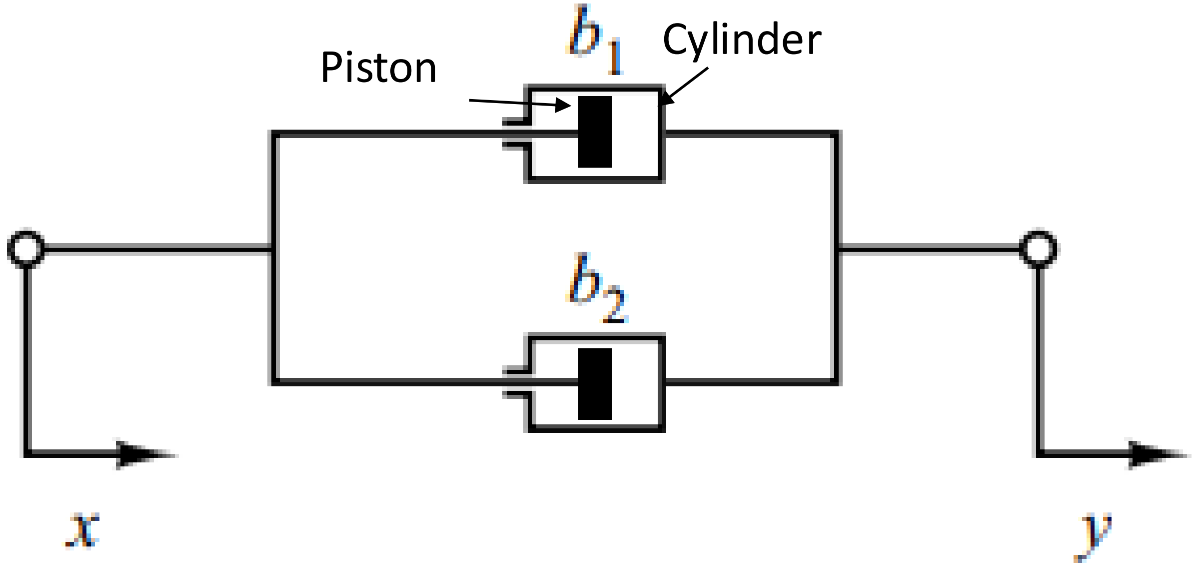


I/P and O/P relationship

- Model dynamic systems in mathematical terms (Why?)
- A set of equations that represents the dynamics of the system
- System may have many mathematical models
- The dynamics of systems may be described in terms of differential equations
- Use physical laws governing a particular system
- Testing a prototype of the device, or measuring its response to inputs
- Deriving reasonable mathematical models is the most important part

Mechanical System Modelling

System consisting of two dampers connected in parallel



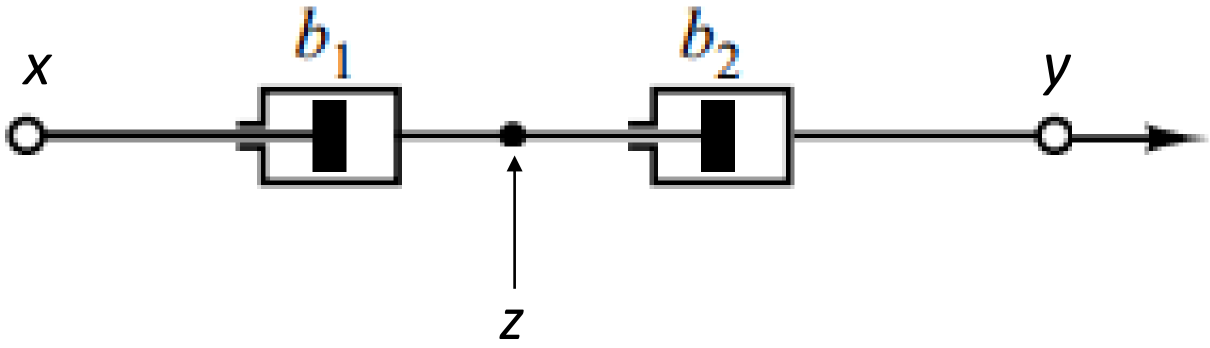
b_1 & b_2 are viscous-friction coefficients

x and y are positions. The force F due to the dampers is

$$F = b_1(\dot{y} - \dot{x}) + b_2(\dot{y} - \dot{x})$$

Exercise: Mechanical System Modelling

System consisting of two dampers connected in series



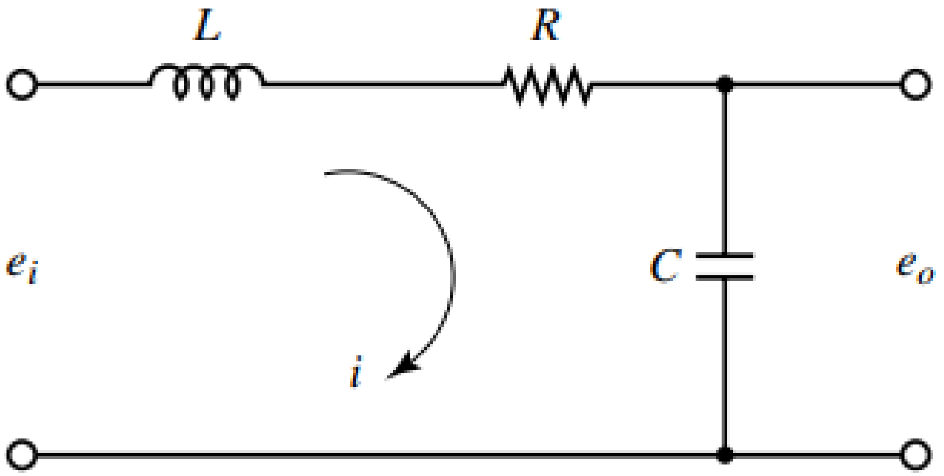
b_1 & b_2 are viscous-friction coefficients

x and y are positions. The force F due to the dampers is

$$F = b_1(\dot{z} - \dot{x}) = b_2(\dot{y} - \dot{z})$$

Electrical System Modelling

Consider the electrical circuit as shown below



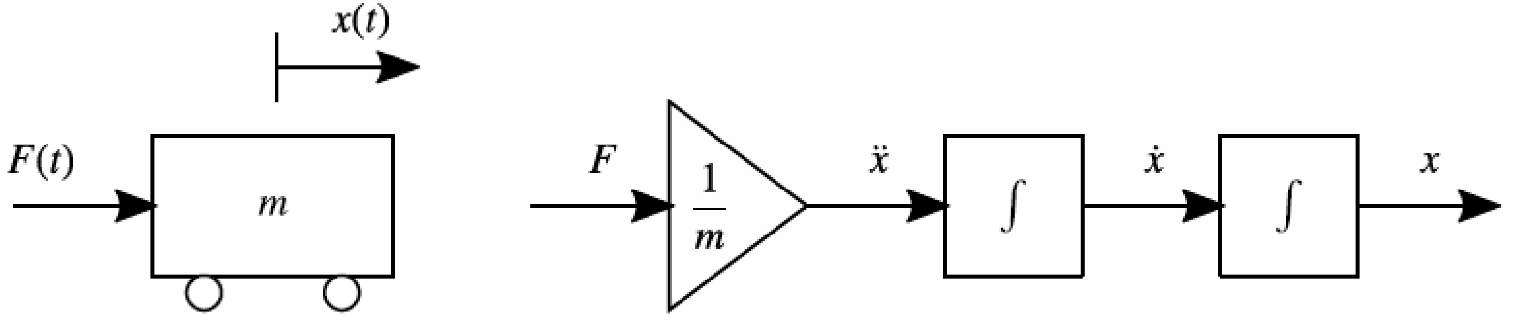
$$L\frac{di}{dt} + Ri + \frac{1}{C}\int i\,dt = e_i$$

A Mechanical System

- A simple trolley is subject to an input force $F(t)$.

- From Newton's 2nd Law, we know
- $F(t) = m\ddot{x}(t)$
- where m is the (constant) mass of the trolley and $\ddot{x}(t)$ is its acceleration, or the second derivative of position. The state/output is then the position $x(t)$.

- As the second derivative of the output is proportional to the input, this is a second-order system.



Describing a Dynamic System

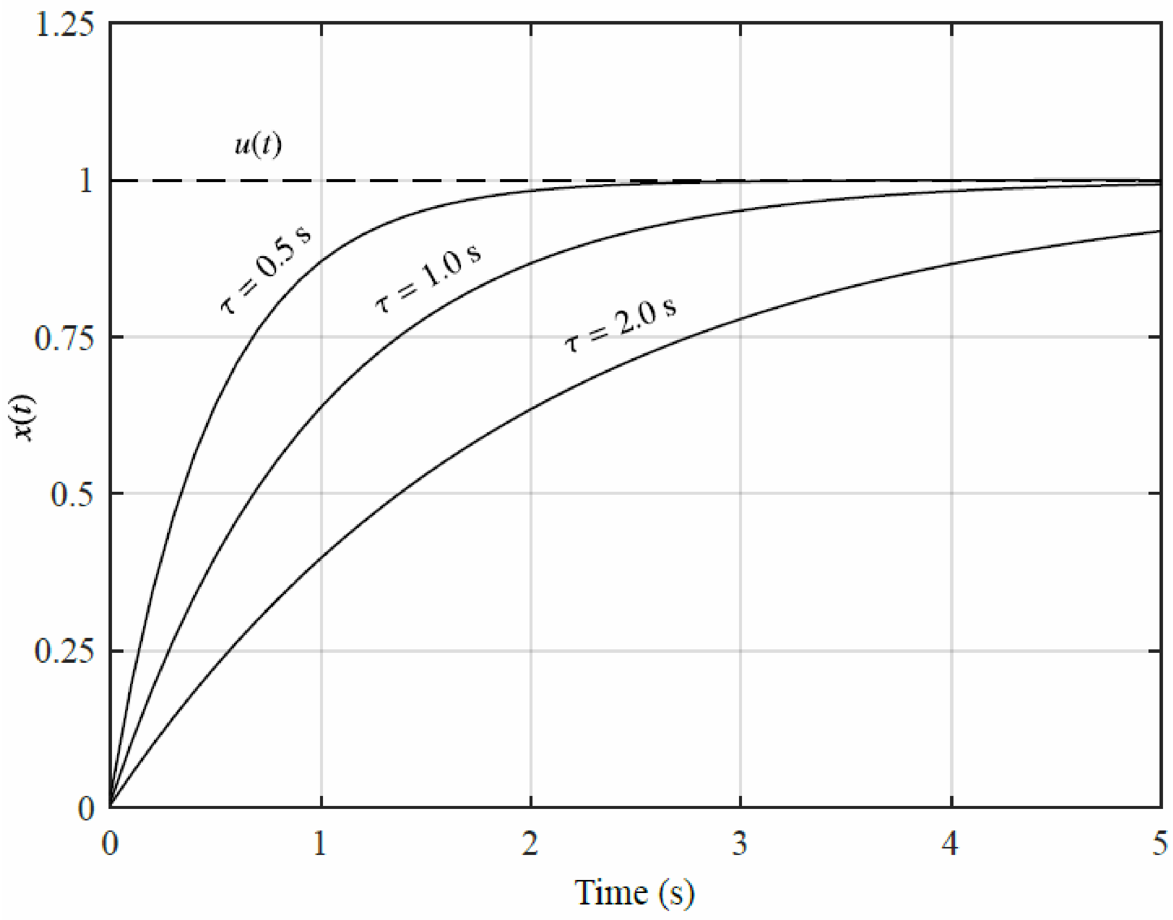
- Equations describing the systems known as system’s equation of motion.
- Describes the O/P of the system
- Describes its derivatives as a function of the I/P
- Can have MIMO

First-Order System

$$\dot{x}(t) + \frac{1}{\tau}x(t) = Ku(t)$$

- where $x(t)$ is the state/output, $u(t)$ is the input, τ is the time constant and K is the input gain.

- Varying τ changes the "speed" of the system, while varying K changes the output magnitude with respect to the input.

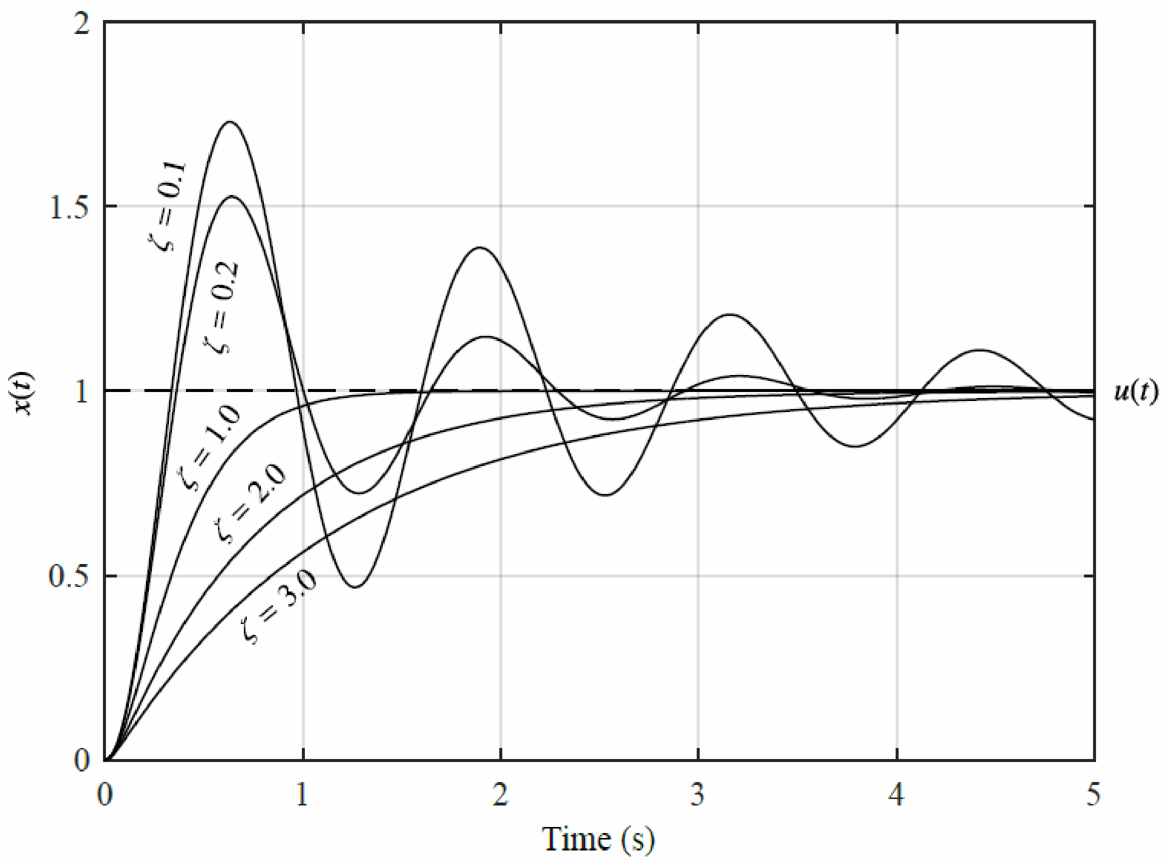


Second-Order System

$$\ddot{x}(t) + 2\zeta\omega_n\dot{x}(t) + \omega_n^2 x(t) = Ku(t)$$

where $x(t)$ is the state/output, $u(t)$ is the input, ζ is the damping ratio, ω_n is the natural frequency and K is the input gain.

Varying ζ changes how much the system oscillates, while varying ω_n changes the "speed" of the system.



Higher-Order Systems

An n^{th} - order system has the general equation of motion

$$x^{(n)}(t) + a_1 x^{(n-1)}(t) + a_2 x^{(n-2)}(t) + \cdots + a_n x(t) = Ku(t)$$

The Laplace Transform

- Differential and integral equations are difficult to model as a block diagram
- The input, output, and system are represented separately with their own equations
- Their interrelationship will be represented as algebraic equations

$$\mathscr{L}[f(t)] = F(s) = \int_{0-}^{\infty} f(t)e^{-st}\,dt$$

$f(t)$	$F(s)$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
$tu(t)$	$\frac{1}{s^2}$
$t^n u(t)$	$\frac{n!}{s^{n+1}}$
$e^{-at}u(t)$	$\frac{1}{s+a}$
$\sin \omega t u(t)$	$\frac{\omega}{s^2+\omega^2}$
$\cos \omega t u(t)$	$\frac{s}{s^2+\omega^2}$

Transfer Functions

- Describe the relationship between the input (I/P) and the output (O/P)

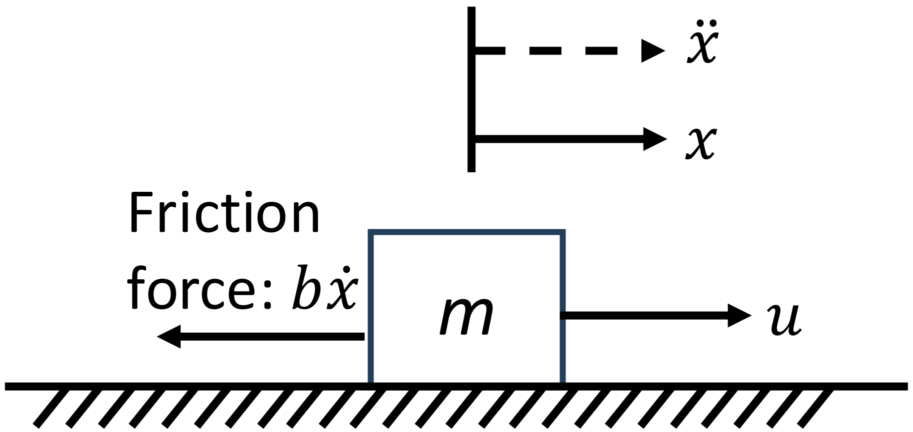
$$\dot{x}(t) + \frac{1}{\tau}x(t) = Ku(t)$$

- Using the Laplace domain, we get

$$sx(s) + \frac{1}{\tau}x(s) = Ku(s)$$

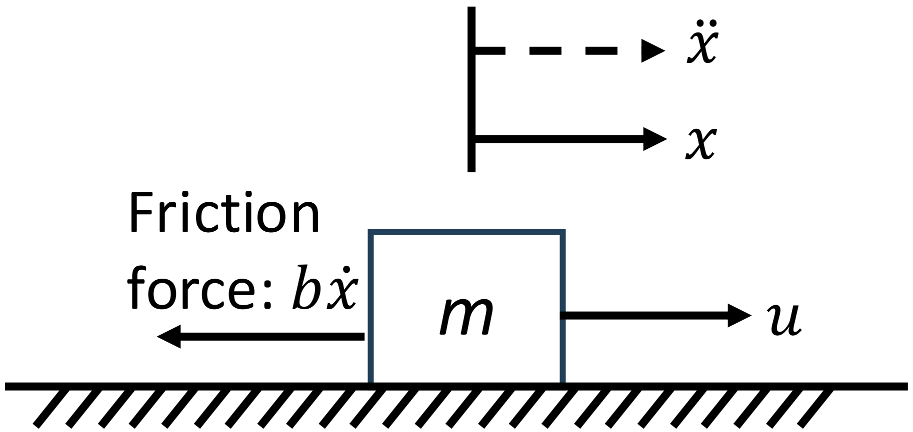
$$\frac{x(s)}{u(x)} = \frac{K}{s + \frac{1}{\tau}}$$

Exercise: Transfer Function for a mechanical system



- Applying Newton's law, $F = m \cdot a$
- F is the vector sum of all forces
- a vector acceleration in m/s^2
- m mass of the body in kg
- A force $u(t)$ is applied to a mass m and an opposing force of friction applies

Exercise: Transfer Function for a mechanical system



- $F = m \cdot a$
- $u - b\dot{x} = m\ddot{x}$
- $\ddot{x} + \frac{b}{m}\dot{x} = \frac{u}{m}$
- Converting using the Laplace transforms
- $s^2X(s) + \frac{b}{m}sX(s) = \frac{1}{m}U(s)$
- $ms^2X(s) + bsX(s) = U(s)$

$$\frac{X(s)}{U(s)} = \frac{1}{ms^2 + bs}$$

Obtain transfer functions for the below systems

Second-order system: $\ddot{x}(t) + 2\zeta\omega_n\dot{x}(t) + \omega_n^2 x(t) = Ku(t)$

n^{th}-order system: $x^{(n)}(t) + a_1 x^{(n-1)}(t) + a_2 x^{(n-2)}(t) + \cdots + a_n x(t) = Ku(t)$

Types of Input

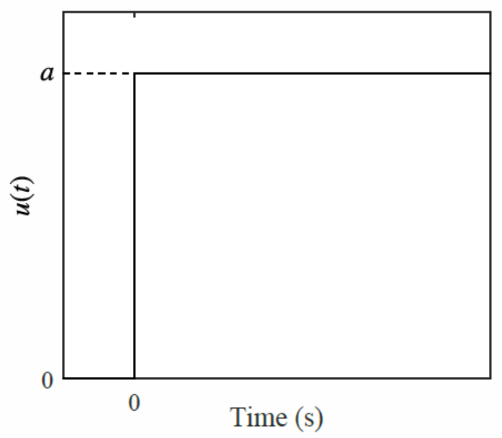
- The objective of a controller is to determine the input $u(t)$ for a specific response.

- However, $u(t)$ can also be a constant or any time-dependent function. By applying a certain input and observing the output response, we can learn much about the system.

The most popular input is the step input, defined by

$u(t) = 0$ for $t < 0$

$u(t) = a$ for $t \geq 0$



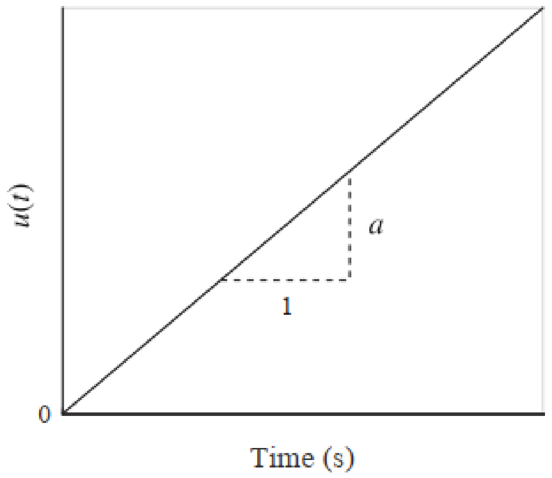
$$\frac{a}{s}$$

Types of Input

The ramp input, defined by

$u(t) = 0$ for $t < 0$

$u(t) = at$ for $t \geq 0$



$$\frac{a}{s^2}$$

Response Properties

- By applying a step input to a system which is at rest, we cause the system to be in forced vibration.

- The system will be in motion before settling to a constant steady-state value. The phase where the system is in motion is called the transient phase. In theory, the steady-state phase occurs at $t = \infty$, although in practice it occurs much sooner.

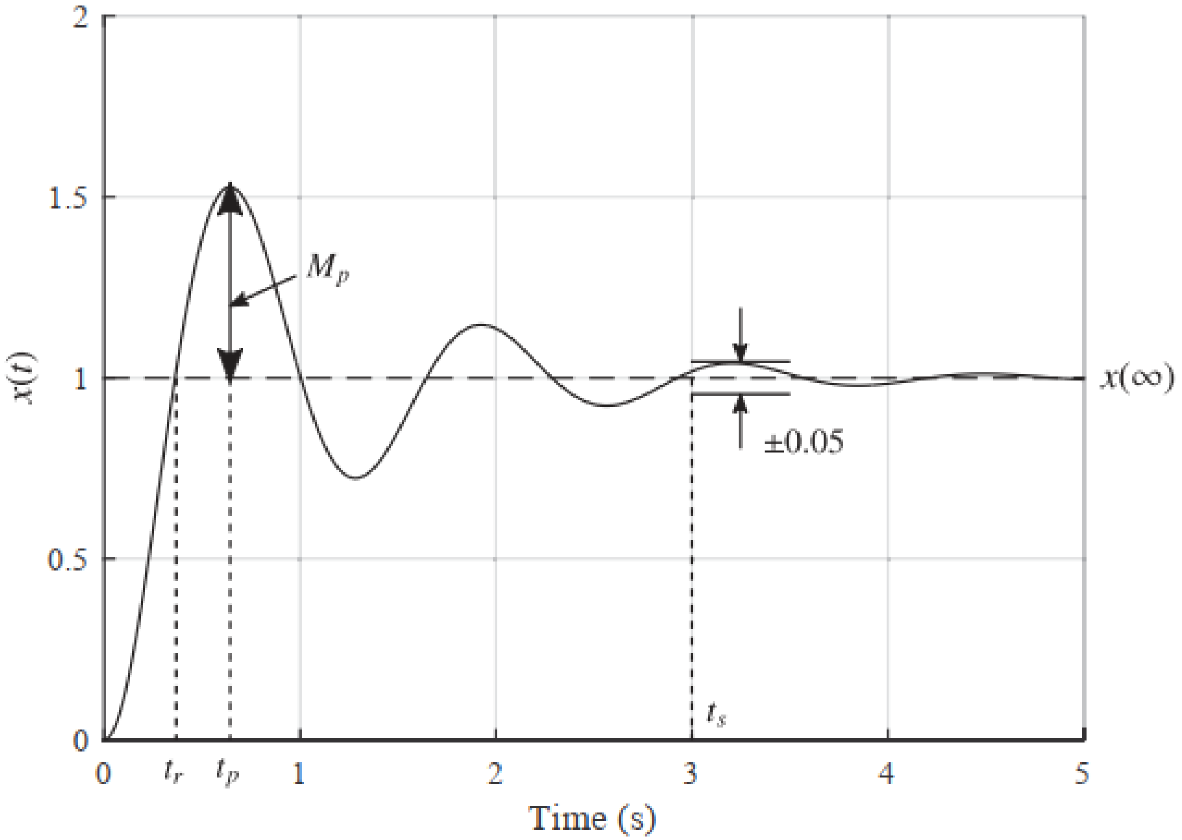
- The steady-state value may be found using the final-value theorem. For a second-order system, the final-value theorem yields

- $x_{ss} = x(\infty) = \frac{Ka}{\omega_n^2}$

Response Properties etc.

The response of a system can be described by several properties

Maximum peak value of the response: M_p
Rise time: t_r
Peak time: t_p
Settling time: t_s



Controlling a System

- These properties only hold if a system is stable. An unstable system will never reach a steady-state value after receiving an input, or will never reach an equilibrium after being disturbed.

- Some systems, such as fighter aircraft, are deliberately unstable. In these cases, we design a control system which stabilises the aircraft.

- We can design our control system so that it gives our complete a system a desired property or set of properties.

Summary of Section 1

- A dynamic system can be set in motion by applying an input or moving the system away from its equilibrium
- A dynamic system can take many forms, with some properties being:
  - The order of the system, $1, 2, 3, \dots n$
  - The stability of the system
  - The shape of the transient response
  - The properties of the steady-state response
- We can use a controller to change the properties of the system by regulating its input
- This means we can:
  - Make an unstable system stable
  - Increase the order of a system
  - Change the damping and speed of a system response

Summary

- Basics of Dynamic Systems
- Mathematical Models
- Transfer Function
- Response Characteristics

References:

-Modern Control Engineering, 5th Edition, K. Ogata

-UESTC3001 2019/20 Notes, J. Le Kernec etc.